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Continuance use of AI coding assistants among South Korean Industry Developers: A survey case study with large language models

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Abstract
Utilizing generative artificial intelligence including large language models (LLMs), into programming tasks has recently garnered significant attention. This paper investigates the motivations determining the adoption of LLM-oriented services, integrating two user-oriented concepts, the technology acceptance model and expectation confirmation theory into the foundational frameworks. By combining these models, the study presents a comprehensive framework for understanding developers' continual usage intentions of these technologies. Analyzing data from 1,338 developers, the study reveals that developers are more inclined to adopt LLM-oriented services for their programming tasks. Key findings include the critical role of perceived ease of use, the alignment of service performance with initial expectations, and the influence of developers' enjoyable feeling on their acceptance. These insights emphasize the importance of ensuring that LLM-oriented services not only meet developers' expectations but also enhance their overall programming experience through ease of use and enjoyment.

Keywords Large language model · Expectation confirmation theory · Technology acceptance model · Programming task · Developers

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1 Introduction

The advent of artificial intelligence (AI) has profoundly impacted and transformed numerous industries, with the information technology sector being one of the most significantly affected (Pantano & Pizzi 2020). Within this sector, programming tasks have experienced substantial changes due to the rapid development of AI-driven technologies, particularly the emergence of large language models (LLMs) such as ChatGPT. These models offer developers advanced tools for code generation, debugging, and learning new programming languages, providing contextual code suggestions, clarifying complex programming concepts, and automating routine coding tasks (Acher et al. 2023; Foundjem et al. 2022; Godliauskas & Šmite 2025; Nam et al. 2024). Despite these apparent benefits, the extent to which these tools are accepted and adopted by developers remains an area requiring further investigation.

Understanding developer acceptance is crucial for the successful deployment and sustained use of AI-driven programming assistants, as developers constitute the primary users of these tools (Wang et al. 2024). While the nascent field of generative AI in software development has seen increasing attention, including studies on LLM adoption (e.g., Russo 2024), there remains a critical need to understand factors driving continual usage intention beyond initial adoption. This study addresses this gap. We aim to establish robust frameworks for analyzing the factors influencing user acceptance by integrating two well-established theoretical models. First, the Expectation Confirmation Theory (ECT) examines how users' expectations, perceived performance, and satisfaction form their continual usage of a technology (Gupta et al. 2021). Second, the Technology Acceptance Model (TAM) utilizes perceived usefulness and ease of use as core motivations of technology acceptance (Park 2024). Our focus on post-adoption dynamics offers a distinct contribution to the literature, moving beyond mere initial acceptance to explore the determinants of continued usage in software engineering workflows.

By integrating ECT and TAM, we aim to offer a comprehensive knowledge of the motivations that form developer acceptance of LLM-oriented services in programming tasks. It aims to elucidate how expectations, performance, satisfaction, and ease of use collectively influence the decision to employ and continue using AI-driven tools in the programming context.

2 Theoretical background

Recent studies on AI-assisted tools in programming tasks, such as GitHub Copilot, examined how developers utilize the tools in their practices. GitHub Copilot introduced by GitHub and OpenAI, is known to improve coding efficiency and to raise challenges associated with code reliability and developer ability (Wermelinger 2023). Several scholars also highlighted that AI-based tools transform the workflow and roles of developers by changing their tasks from code production to code review and optimization (Cheng et al. 2024). Moreover, Tabnine and Cursor AI, two well-known AI-driven code writing tools, present well-defined code generation and debugging functions, further reducing developers' cognitive workload and improving their productivity (Krishna & Meda 2024). Although they present the notable

benefits of AI-assisted programming tasks, they rarely present developers' motivations for employing LLM-oriented services in programming tasks.

The rapid advancements of large language models (LLMs) impact the software development landscape, with tools such as GitHub Copilot and ChatGPT rapidly integrating into developer workflows. These AI-powered assistants promise to revolutionize coding practices, from accelerating code generation and debugging to aiding in documentation and refactoring tasks (Chen 2025; Chen et al. 2023). Although academic research on technology adoption models provides foundational insights, the specific presentations of LLM integration into software engineering workflows warrant focused investigation. Practitioner feedback and industry reports, such as those from major developer surveys, consistently highlight both the excitement and the challenges associated with these tools, pointing towards a complex interplay of perceived benefits and practical considerations in their actual use (Aguiar et al. 2024; Chen et al. 2025). Despite their growing utilizations within the practitioner community, a comprehensive empirical understanding of the factors driving developers' continual usage intention of LLM-oriented programming tools, especially through established theoretical lenses, remains underdeveloped. This gap is crucial to address as it directly impacts effective tool design, organizational adoption strategies, and developer productivity metrics commonly tracked in the industry.

Thus, after presenting the overview of ECT and TAM, we attempt to provide a developer-centric perspective toward his/her adoption with LLM-oriented services in programming tasks.

2.1 Expectation Confirmation Theory (ECT)

ECT posits that users' continual employment of a specific technology is affected by the confirmatory level of their beginning expectations and the resulting level of satisfaction (Brown et al. 2012). Originally developed to understand consumer behavior, ECT are widely applied in technology acceptance research. According to ECT, the process begins with users forming initial expectations about a technology (Thong et al. 2006). After engaging with the technology, users evaluate its performance against these expectations. When the technology's performance meets or exceeds expectations, confirmation occurs, leading to increased satisfaction (Sharma et al. 2024). Higher satisfaction subsequently enhances the intention to continue using the technology. ECT is typically structured around four key factors: expectations, perceived performance, confirmation, and perceived satisfaction (Oliver 1980).

2.2 Technology Acceptance Model (TAM)

TAM is one of the most influential frameworks for understanding user behavior in technology adoption. TAM is grounded in two foundational theories: the Theory of Reasoned Action (TRA) and the Theory of Planned Behavior (TPB). TRA, introduced by Ajzen and Fishbein (1975), asserts that "*an individual's behavior is primarily determined by their intention to perform the behavior, which is influenced by their attitudes and subjective norms*". In TAM, both an individual's perceived usefulness and ease of use shape his/her attitudinal minds while primarily examining their willingness to engage with a technology. TPB examines TRA by incorporating the concept of perceived behavioral control to explain how external factors affect intention and behavior (Ajzen 1991). While TAM does not explicitly include

the behavioral control, it recognizes that external factors, such as service design and user training, can affect users' feelings of usefulness and ease of use.

After its introduction by Davis (1989), two primary factors—perceived usefulness and ease of use— were highlighted as core motivations of users' attitudes toward employing a particular technology. These motivations, along with the user's attitude, primarily predict his/her intention and actual usage behavior. TAM has been widely employed and adapted across various domains, incorporating additional factors such as perceived enjoyment and facilitating conditions.

2.3 Extension of ECT, TAM and others

The integration of ECT and TAM presents a multi-dimensional framework for improving our knowledge on both the initial acceptance and continued use of technology. ECT offers insights into how users' satisfaction and intentions for continued use are shaped by their initial expectations and subsequent experiences (Oh et al. 2022). Meanwhile, TAM significantly highlights both users' perceived usefulness and ease of use in forming their attitudes toward and intentions to use a specific technology (Han & Park 2021).

Moreover, Russo (2024) examined generative AI adoption in software engineering by utilizing TAM, diffusion of innovation theory with several key motivations that affect the initial adoption of generative AI tools. Moreover, Chen (2025) employed ECT to evaluate how optimism forms users' intention to consistently use ChatGPT, providing some highlights into how their expectations affect consistent engagement. Furthermore, Zheng et al. (2024) integrated ECT and TAM to explore user loyalty for generative AI, examining general adoption behaviors rather than specific domain-oriented cases. While they provide several insights, they mainly examine general AI adoption trends. In contrast, we focus on developers' continual intention to use LLM-oriented programming tools within software engineering environments. By analyzing LLM adoption in practical programming tasks, we extend prior research by addressing how expectation confirmation, usability, and satisfaction influence developers' engagement with existing workflows. Furthermore, we provide new empirical findings by examining the role of perceived cost on developers' continued use of LLMs, challenging conventional assumptions about cost sensitivity in technology adoption.

The proposed integrated model posits the following: (1) Developers form initial expectations regarding the usefulness and ease of use of AI-driven programming assistants; (2) After interacting with these assistants, they assess performance relative to their expectations; (3) When expectations are met, satisfaction increases, reinforcing a positive attitude toward the technology; (4) This satisfaction, influenced mainly by perceived usefulness and ease of use, determines the intention to continually utilize the technology. Based on these principles, we introduce our integrated model with corresponding hypotheses.

3 Research model and hypotheses

We present a research model, which is examined by the integration of ECT and TAM to capture the factors affecting the acceptance of LLM-oriented services in programming tasks. The proposed model includes the following constructs: Expectation, Perceived Usefulness,

Perceived Ease of Use, Confirmation, Satisfaction, Perceived Cost, and Continual Intention to Use.

3.1 Role of expectation

In the context of information systems, expectation is defined as "*an individual's evaluation of a specific service based on the information provided before use*" (Ngah et al. 2023). Users' expectations shape their initial perceptions and conceptual satisfaction. Generally, expectations are influenced by prior knowledge, observations, media exposure, and the experiences of others (Heit 1994). For example, if a user expects a service to provide significant, coherent, and effective responses, their expectations will subsequently be confirmed or disconfirmed based on their experience with the service (Hsieh & Yuan 2021).

Regarding LLM-oriented services in programming tasks, users evaluate these services against their initial expectations, which then shapes their satisfaction levels. For example, Reychav et al. (2021) analyzed 190 samples of mobile technology use and found that perceived expectations significantly influenced confirmation ($\beta = 0.41$, $p < 0.001$). Similarly, Wu et al. (2024) investigated 459 respondents' continual intention to use mobile health applications and reported a positive association between expectation and confirmation ($\beta = 0.478$, $p < 0.001$). Additionally, Hsiao (2018) demonstrated that users' expectations directly influence satisfaction with online stores ($\beta = 0.27$, $p < 0.05$). Thus, considering the results of prior literature on expectation, the following hypotheses are presented:

H1: A higher expectation of LLM-oriented services in programming tasks leads to a higher level of confirmation of the services.

H2: A higher expectation of LLM-oriented services in programming tasks leads to a higher level of satisfaction with the services.

3.2 Role of confirmation

Confirmation occurs when there is alignment between users' expectations and their perceived actual performance of a service. This alignment can significantly affect user satisfaction, as users tend to be more satisfied when the service meets or exceeds their expectations (Kim et al. 2021). For instance, if a user finds that an LLM-oriented service provides clear, consistent, efficient, and concise results, their satisfaction with the interaction can be improved.

Confirmation also influences users' perceived ease of use and usefulness of LLM-oriented services. These two factors, which are central to TAM, are shaped by users' experiences and their confirmation of expectations during their interactions with LLM-oriented services. Sørebø and Eikebrokk (2008) found that confirmation directly determines ease of use ($\beta = 0.56$, $p < 0.001$), usefulness ($\beta = 0.36$, $p < 0.001$), and satisfaction ($\beta = 0.40$, $p < 0.001$). Similarly, Joo and Choi (2016) showed that online service users' feelings of confirmation directly influence their usefulness feeling ($\beta = 0.449$, $p < 0.01$) and their satisfaction level ($\beta = 0.300$, $p < 0.01$), and indirectly affect their continual usage intention. Hsiao (2018) and also highlighted the notable role of confirmation in examining satisfaction and usage intention. Thus, with consideration of these findings, the following hypotheses are presented:

H3: A greater level of confirmation of LLM-oriented services in programming tasks leads to a higher level of satisfaction with the services.

H4: A greater level of confirmation of LLM-oriented services in programming tasks leads to a higher perceived ease of use of the services.

H5: A greater level of confirmation of LLM-oriented services in programming tasks leads to a higher perceived usefulness of the services.

3.3 Role of ease of use

Perceived ease of use is defined as "*the degree to which a prospective user expects the target system to be free of effort*" (Davis 1989; Fraccascia & Nastasi 2023). When users think an LLM-oriented service easy to use, it positively influences their perceived satisfaction with the service. In workplace information systems, several scholars validated a positive connection between perceived satisfaction and ease of use.

For example, Park (2020) demonstrated that the ease of use is a decisive motivation of satisfaction with smart wearable devices, based on 1,380 valid samples (Critical ratio = 2.293; $p < 0.05$). Hahm et al. (2022) also found that perceived ease of use in remote working environments influences satisfaction levels (based on 124 respondents). Thus, with consideration of these findings, the following hypothesis is presented:

H6: A greater level of perceived ease of use of LLM-oriented services in programming tasks leads to a higher level of satisfaction with the services.

3.4 Role of perceived usefulness

The general meaning of perceived usefulness is "*the degree to which an individual believes that using a particular system would enhance their overall job performance*" (Davis 1989; Ahn & Park 2023). Related to LLMs, it refers to "*the extent to which users believe that the LLM can effectively help them achieve their goals, whether those goals involve generating content, solving problems, or enhancing productivity*" (Davis 1989; Park 2024). When users perceive LLM-oriented services in programming tasks as useful, they are likely to experience enhanced satisfaction and a greater intention to continually employ the services.

Akdim et al. (2022) examined that perceived usefulness significantly predicts overall satisfaction with social mobile applications ($\beta = 0.285$, $p < 0.001$). Al-Fraihat et al. (2020) similarly validated that perceived usefulness plays a motivational factor of satisfaction with online education systems ($\beta = 0.277$, $p < 0.001$). Park (2020) further showed that perceived usefulness positively influences users' continual intention to employ smart wearable devices ($\beta = 0.120$, $p < 0.001$). Thus, with consideration of these findings, the following hypotheses are presented:

H7: A higher level of perceived usefulness of LLM-oriented services in programming tasks leads to a higher level of satisfaction with the services.

H8: A higher level of perceived usefulness of LLM-oriented services in programming tasks leads to a greater level of continual intention to use the services.

3.5 Role of satisfaction

Related to information services and systems, user satisfaction is mainly defined as "*the feeling of pleasure or displeasure resulting from aggregating all the benefits that a person hopes to receive from interaction with an information system*" (Abdennebi 2023). When

users are generally satisfied with their experience using an LLM-oriented service, they tend to continue using it. Satisfaction reflects a positive emotional response when a service meets user expectations.

Chiu et al. (2007) found that satisfaction plays a significant motivation in determining users' continual intention to use online learning environments ($\beta=0.86$, $p<0.05$). Similarly, Zhonggen and Xiaozhi (2019) showed that satisfaction strongly predicts continued intention to use mobile learning technologies ($\beta=0.31$, $p<0.01$). Thus, with consideration of these findings, the following hypothesis is presented:

H9: A higher level of satisfaction with LLM-oriented services in programming tasks leads to a greater level of continual intention to use the services.

3.6 Role of perceived cost

Perceived cost is examined as "*the extent to which an individual believes that using a specific system is costly*" (Marzouk et al. 2024). Users are more likely to continue using an LLM-oriented service if they perceive the cost as affordable relative to the benefits received. For instance, Li et al. (2023) presented that perceived cost is a crucial obstacle to the positive intention to use financial information services ($\beta=-0.580$, $p<0.001$). Chan and Zhou (2023) also identified a negative correlation between perceived cost and intention to use generative artificial intelligence-oriented technologies (based on 405 respondents). Thus, with consideration of these findings, the following hypothesis is presented:

H10: A lower level of perceived cost of LLM-oriented services in programming tasks leads to a greater level of continual intention to use the services.

3.7 Research model

Considering the proposed hypotheses, our integrated model is examined (Fig. 1).

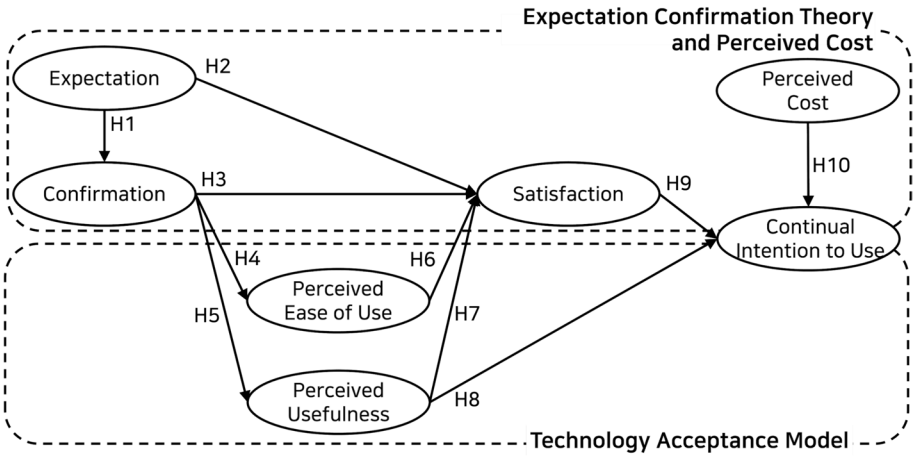

Fig. 1 Proposed model

4 Methodology

The research employed a survey methodology to collect data on users' acceptance of LLM-oriented services in programming tasks. The development of the survey instrument followed guidelines from prior validated research (Park 2024; Yoo et al. 2020).

A total of 29 questionnaire items were compiled from previous user-centric research to ensure reliability and validity. The questions in this study were translated from English to Korean using a back-translation process, where bilingual researchers independently translated the items and checked for discrepancies in meaning. Moreover, a panel of three experts (communication, artificial intelligence, and information systems) reviewed the translated items to ensure conceptual clarity and relevance within the Korean context.

A pilot survey was examined with a group of 20 graduate students, who had more than 2 years of experience in programming tasks, majoring in computer science, artificial intelligence, data science and engineering, to evaluate the comprehensibility and relevance of the translated items. Examining the feedback from the pilot survey, 22 questionnaire items were retained, and five items were excluded due to redundancy, ambiguity, or irrelevance. The final set of items employed in the survey is presented in Appendix A (Table 9).

4.1 Data collection

A South Korean professional survey company conducted an online-based survey within three months in 2024 (from April 2nd to July 1 st, 2024). It recruited 1,994 South Korean developers as our respondents. All participants in the main survey were asked to answer each questionnaire item using a 7-point Likert scale (1: Disagree ~ 7: Agree, 4: Neutral). Then, we collected 1,521 responses (76.3% response rate). After excluding the developers with no usage experience or a skipped response on LLM-oriented services or less than 90-day experience on the services, 1,338 validated responses were employed in our analysis. Moreover, all respondents in the survey were South Koreans. Table 1 examines their demographic information in the main survey.

Table 1 Respondents' demographic information (N = 1,338)

Age	n (%)	Living areas	n (%)
21 ~ 30	392 (29.3%)	Small/medium-sized cities	197 (14.7%)
31 ~ 40	815 (60.9%)	Rural	94 (7.0%)
41 ~ 50	86 (6.4%)	Metropolis	1,047 (78.3%)
51 ~ 60	30 (2.2%)		
Above 60	15 (1.1%)		
Working ranks	n (%)	Period of using experience	n (%)
Staff	148 (11.1%)	3 ~ 12 months	204 (15.2%)
Senior staff ~ Assistant Manager	1,082 (80.9%)	12 ~ 18 months	332 (24.8%)
Manager ~ General Manager	75 (5.6%)	18 ~ 24 months	691 (51.6%)
Director or above director	33 (2.5%)	Above 24 months	111 (8.3%)
Gender	n (%)		
Female	277 (20.7%)		
Male	1,061 (79.3%)		

4.2 Focus group interviews

To further validate the findings derived from our quantitative survey and ensure their practical relevance, we conducted a series of focus group interviews with 10 professional developers. These participants were distinct from the main survey sample but shared similar demographic and professional characteristics, including experience with LLM-oriented programming tools. The interviews were semi-structured, lasting approximately 30 min each, and aimed at exploring developers' real-world experiences with LLM tools, their perceptions of usefulness, ease of use, and factors contributing to their satisfaction and continued usage. Key themes and opinions emerging from these qualitative discussions were systematically analysed to identify points of convergence and divergence with our quantitative model results, thereby serving as a critical step in triangulating our findings with practitioner insights.

5 Findings

5.1 Descriptive analysis, validity and reliability tests

Table 2 examines the descriptive results. Moreover, we employed a structural equation modelling method, as well as confirmatory factor analysis. For the internal, convergent and discriminant reliabilities, the following guidelines were required (Collier 2020; Kline 2023).

- Factor loadings: Higher than 0.7.
- Cronbach's alpha: Higher than 0.7.
- Average variance extracted: Higher than 0.5
- Correlation degrees between two specific constructs: Greater than each square root degree of the average variance extracted levels
- Composite reliability: Higher than 0.7.

All criteria were met in our analysis, as shown in Tables 3 and 4, presenting the reliability of the measurement model.

We assessed potential collinearity issues by examining the variance inflation factor (VIF) values and evaluated common method bias (CMB) using Harman's single-factor test (Marcoulides and Raykov 2019). The VIF values for all independent constructs ranged from 1.99 to 2.84, well below the commonly accepted threshold of 5, indicating no multicollinearity

Table 2 Mean and standard deviation of the employed factors

Construct	Mean (Standard deviation)	Construct	Mean (Standard deviation)
Expectation	5.11 (0.94)	Confirmation	5.01 (1.01)
Perceived Ease of Use	5.02 (0.97)	Perceived Usefulness	5.00 (1.00)
Satisfaction	5.09 (1.04)	Perceived Cost	2.98 (0.99)
Continual Intention to Use	5.03 (1.01)		

Table 3 Internal reliability and convergent validity

Construct	Cron-bach's alpha	Factor loading	Composite reliability	Average variance extracted
Expectation	0.846	0.810	0.896	0.837
		0.800		
		0.842		
		0.855		
Confirmation	0.838	0.849	0.880	0.710
		0.866		
		0.812		
Perceived Ease of Use	0.810	0.781	0.850	0.654
		0.809		
		0.835		
Perceived Usefulness	0.795	0.732	0.882	0.715
		0.894		
		0.900		
Satisfaction	0.880	0.878	0.910	0.771
		0.910		
		0.845		
Perceived Cost	0.890	0.848	0.887	0.724
		0.850		
		0.829		
Continual Intention to Use	0.912	0.874	0.819	0.602
		0.860		
		0.811		

Table 4 Summary of discriminant validity

Construct	1	2	3	4	5	6	7
1. Expectation	0.915						
2. Confirmation	0.442	0.843					
3. Perceived Ease of Use	0.309	0.232	0.809				
4. Perceived Usefulness	0.294	0.196	0.304	0.846			
5. Satisfaction	0.341	0.609	0.331	0.252	0.878		
6. Perceived Cost	−0.322	−0.192	−0.441	−0.155	−0.417	0.851	
7. Continual Intention to Use	0.501	0.594	0.402	0.351	0.256	−0.411	0.776

concerns. Furthermore, the first factor accounted for 29.93% of the variance, which is below the recommended 50% threshold (Podsakoff et al. 2003), suggesting that there is no common method bias issue.

5.2 Fit indices

Table 5 shows the computed fit indices of the models. All indices indicate an acceptable fit according to the recommendations from prior research.

Table 5 Summary of fit indices

Indices	Measurement model	Research model	Satisfactory levels
Comparative fit indices	0.904	0.901	>0.900
Adjusted Goodness-of-fit indices	0.902	0.902	>0.900
Incremental fit indices	0.911	0.911	>0.900
Normed fit indices	0.902	0.901	>0.900
Standardarzd root mean square residual	0.050	0.050	<0.070
Root mean square error of approximation	0.086	0.085	<0.090
Goodness-of-fit indices	0.904	0.904	>0.900

5.3 Results of testing hypotheses

For the structural model, we examine our employed hypotheses using maximum Likelihood estimation approaches. Standard path coefficients with their significance are evaluated via 5,000 resampling bootstrapping procedures. Figure 2 and Table 6 summarize the results of the presented. The developers' continual intention to use the technology ($R^2=0.752$) is primarily influenced by their usefulness feeling (H8: $\beta=0.573$, $p<0.001$) and satisfaction (H9: $\beta=0.330$, $p<0.001$). However, no relationship is observed between perceived cost and intention to use (H10: $\beta=0.039$, $p=0.215$).'

Three factors significantly affect developers' perceived satisfaction ($R^2=0.683$): perceived usefulness (H7: $\beta=0.218$, $p<0.001$), ease of use (H6: $\beta=0.190$, $p<0.001$), and confirmation (H3: $\beta=0.442$, $p<0.001$). No relationship is found between expectation and perceived satisfaction (H2: $\beta=0.053$, $p=0.193$).

Confirmation emerges as a core determinant of perceived ease of use (H4: $\beta=0.803$, $p<0.001$, $R^2=0.644$) and usefulness (H5: $\beta=0.805$, $p<0.001$, $R^2=0.647$). Moreover, developers' expectations are significantly related to their level of confirmation (H1: $\beta=0.709$, $p<0.001$, $R^2=0.503$).

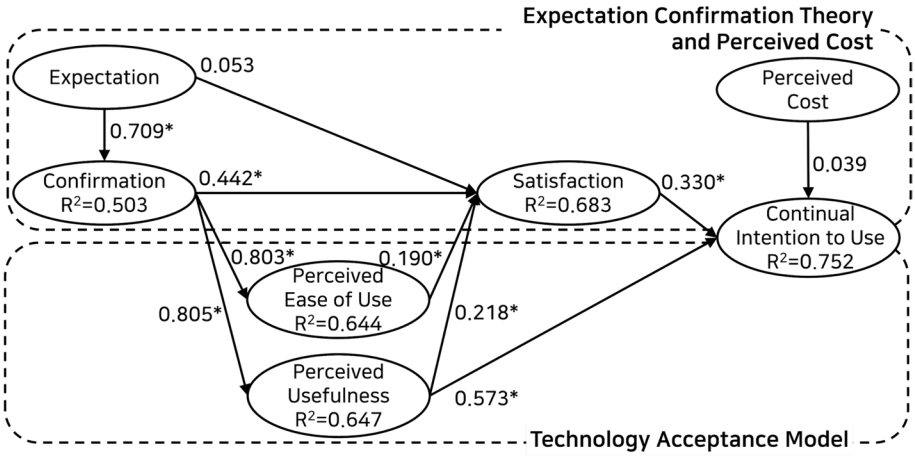

Fig. 2 Results of the research model; *$p<0.001$

Table 6 Summary of testing the proposed hypotheses; EXP: Expectation, CON: Confirmation, PU: Perceived Usefulness, PC: Perceived Cost, SAT: Satisfaction, PEOU: Perceived Ease of Use, CIU: Continual Intention to Use, CR: Critical Ratio, *$p<0.001$

Hypothesis	Standard Coefficient (CR)	Standard error	Results
H1. EXP CON	0.709 (17.362*)	0.028	Supported
H2. EXP SAT	0.053 (1.303)	0.040	Rejected
H3. CON SAT	0.442 (4.262*)	0.046	Supported
H4. CON PEOU	0.803 (17.706*)	0.059	Supported
H5. CON PU	0.805 (18.691*)	0.063	Supported
H6. PEOU SAT	0.190 (3.304*)	0.062	Supported
H7. PU SAT	0.218 (3.933*)	0.054	Supported
H8. PU CIU	0.573 (12.694*)	0.043	Supported
H9. SAT CIU	0.330 (7.609*)	0.042	Supported
H10. PC CIU	0.039 (1.241)	0.042	Rejected

Regarding the total effects of the employed constructs influencing developers' continual intention to use, confirmation has the strongest impact (0.716). Among the constructs derived from the TAM, perceived usefulness exerts a greater impact on the intention (0.646) compared to perceived ease of use (0.063). Furthermore, both expectation (0.525) and satisfaction (0.330) demonstrate moderate impacts on the intention to continually employ the technology (Table 7). We also computed Stone-Geisser's Q^2 values for the predictive relevance of our research model. The levels for the dependent constructs ranged from 0.301 to 0.514 (Table 8), presenting suitable predictive relevance.

We conducted a series of multivariate analysis of variance (MANOVA) to present whether developers' demographic characteristics (e.g., living areas) influence the employed constructs in the research model. We found no statistical impacts on the constructs, indicating that the characteristics did not relate to their perceptions.

5.4 Qualitative insights from Focus Group interviews

To complement our quantitative findings and provide a richer, practitioner-centric understanding of LLM adoption, we conducted focus group interviews with 10 developers. The insights gleaned from these sessions largely corroborated the significant relationships identified in our structural model, reinforcing the empirical grounding of our theoretical framework.

Table 7 Total standardized effects of the employed constructs on the intention

Constructs	Expectation	Confirmation	Perceived Ease of Use	Perceived Usefulness	Satisfaction	Perceived Cost
Indirect	0.525	0.716	-	0.073	-	-
Direct	-	-	0.063	0.573	0.330	0.039
Total Effects	0.525	0.716	0.063	0.646	0.330	0.039

Table 8 Stone-Geisser's Q^2 values for our predictive relevance

Constructs	Continual Intention to Use	Satisfaction	Perceived Usefulness	Perceived Ease of Use	Confirmation
Q^2 values	0.514	0.441	0.398	0.390	0.301

Consistent with our quantitative finding that Perceived Usefulness significantly influences Continual Usage Intention, developers frequently articulated how LLM tools enhanced their work efficiency and quality. For example, *Developer 4* commented, '*My tool speeds up writing codes. I am no longer spending time on mundane syntax; I can focus on the logic*.' Similarly, *Developer 7* highlighted its role in problem-solving: *'When I'm stuck, asking the LLM for different approaches or code snippets is quicker than searching other websites and often more tailored*.' These statements underscore the practical utility of LLMs in accelerating development tasks and improving solution quality, directly supporting our usefulness construct.

The importance of Perceived Ease of Use, as shown in our model, was also echoed in the interviews. Developers emphasized the intuitive nature and seamless integration of these tools. *Developer 2* noted, '*The best part is how it just works within a series of IDEs. I do not have to switch contexts or learn a complex new interface*.' *Developer 9* added, *'It felt natural to start using; the learning curve was almost flat for basic suggestions*.' These qualitative insights affirm that a low barrier to entry and smooth workflow integration are critical for initial and continued acceptance, aligning with our quantitative results.

Our finding that Satisfaction is a strong predictor of Continual Usage Intention was deeply resonated with by the focus group participants. Satisfaction often stemmed from the tool's ability to meet or exceed expectations. *Developer 1* expressed, '*I am pleased when it generates perfectly accurate code on the small number of attempts*.' This comment illustrates how the alignment of service performance with user expectations directly contributes to a high level of satisfaction, which, in turn, drives their intention to continue using the tools.

These qualitative insights from the focus group interviews provide empirical richness and contextual depth, triangulating our statistical findings and strengthening the overall validity and practical relevance of our research on developer acceptance of LLM-oriented programming tools.

6 Discussion

This study examines the motivational factors affecting developers' acceptance of LLM-oriented services in programming tasks. Integrating the ECT and TAM proposes a comprehensive framework that includes perceived usefulness, expectation, confirmation, ease of use, satisfaction, and cost to understand developers' continual intention to use these services.

From the 1,338 validated responses, the findings reveal that developers' initial expectations significantly affect their confirmation, which subsequently influences their usefulness feeling and ease of use. Satisfaction, primarily driven by perceived usefulness, emerges as a core motivation of developers' continual intention to use LLM-oriented services.

Our empirical findings contribute to the broader discourse on AI adoption by both validating and differentiating from existing models, particularly those exploring generative AI in software development (e.g., Russo 2024). The prominence of perceived usefulness in our model for continual usage intention underscores its critical role within the specific context of LLM-oriented programming tools. This particular emphasis validates the continued relevance of core acceptance constructs within a rapidly evolving technological landscape, while also highlighting the practical implications for tool developers focusing on user expe-

rience in coding environments. Furthermore, our focus on the ECT provides a unique lens to understand how developers' post-use evaluations, specifically satisfaction with the tool's performance relative to their initial expectations, significantly drive their decision for continued engagement. This dynamic perspective extends beyond static adoption models, offering actionable insights for fostering long-term adoption and integration of LLMs into daily software engineering practices.

Our findings can also contribute to software engineering contexts by indicating the role of expectation confirmation in the continual use of LLMs. Prior research focused on initial adoption behaviors of AI-assisted tools, presenting perceived usefulness and ease of use as key determinants (Russo 2024). However, we extend this statement by presenting that expectation confirmation significantly shapes both the ease of use and usefulness, reinforcing the importance of aligning LLM performance with user expectations for long-term engagement. Furthermore, we challenge existing insights on cost sensitivity in AI adoption. Whereas prior technology adoption research suggested that perceived cost is a major barrier to adoption (Chen 2025), the findings of this study indicate that cost does not significantly impact developers' continued use of LLMs. This suggests that for developers, productivity gains outweigh financial concerns, marking a key distinction from prior studies that examined broader AI adoption patterns. By integrating ECT and TAM within the context of LLM adoption behavior, we provide a developer-specific perspective on how expectation-driven satisfaction sustains long-term engagement, contributing to filling a gap in existing software engineering research.

Interestingly, perceived cost does not significantly impact the intention to use these services, suggesting that other positive factors in the research model outweigh concerns about affordability. These results underscore the importance of aligning the performance and usability of LLM-oriented services with developer expectations to encourage long-term adoption. The practical implications present that service providers should shed light on the benefits and user-friendly aspects of their services to promote widespread acceptance.

More specifically, the study finds that the confirmation of expectations plays a crucial role in shaping both perceived ease of use and usefulness, core factors of TAM. Developers' initial expectations significantly influence their subsequent confirmation, suggesting that their beliefs about LLMs' capabilities strongly impact their acceptance (H1). However, contrary to initial assumptions, expectations do not directly affect satisfaction (H2). This finding indicates that while expectations set the stage for evaluating LLMs, it is the actual performance and alignment with those expectations (confirmation) that primarily drive satisfaction (H3). This aligns with ECT, where satisfaction is a result of confirmed expectations, emphasizing that developers reassess their expectations based on real-world experiences with LLMs.

Moreover, the roles of ease of use and usefulness are validated within the integrated framework. Both constructs are significant determinants of developers' satisfaction (H6 and H7) and their continual intention to use (H8). Developers perceive these services as valuable tools that enhance productivity and ease the completion of programming tasks, directly contributing to satisfaction and fostering sustained usage. The strong relationship between perceived usefulness and continual intention (H8) emphasizes that developers are highly motivated to engage with these services when they recognize their significant value in enhancing daily programming tasks. These findings align with previous research on technology acceptance (Padiyath et al. 2024; Rasnayaka et al. 2024), where perceived usefulness consistently emerges as a robust predictor of technology adoption and sustained use.

It also indicates that developers prioritize tangible benefits when they decide to employ LLM-oriented tools. Satisfaction, while also significant, plays a notable role, as it addresses how well the tool aligns with developers' prior expectations.

In particular, based on these results, developers are required to utilize the tools for specific coding tasks where their benefits are most evident, such as code generation, error detection, and documentation assistance, to maximize their effectiveness (H8). Furthermore, considering the notable relationship between perceived ease of use and satisfaction (H6), developers can take several customization options and functions that improve their workflow usability.

Moreover, based on the relationships between expectation confirmation and perceived ease of use with usefulness (H4 and H5), the results provide some pieces of evidence that designers are required to align user expectations on the performance by providing guidance and suggestions which can improve over time. Moreover, it highlights that developers are more likely to consistently use the tools when their beginning expectations align with actual experiences, enhancing the significance of building realistic expectations in AI-assisted software development. Considering a software engineering context, these two factors may influence measurable quality metrics. For instance, developers who perceive LLMs as useful may feel efficiency gains through reduced effort. However, misalignment between expectations and actual tool performance may cause increased debugging time, affecting overall task productivity. Thus, future research should examine how expectation confirmation and satisfaction relate to objective software quality measures, such as code maintainability, defect rates, and efficiency improvements.

Interestingly, while it was hypothesized that perceived cost would negatively influence developers' intention to use the services (H10), no significant association was observed. This suggests that the benefits and utilities provided by the services may outweigh cost concerns, particularly when perceived usefulness is high. Developers may view the services as essential tools that justify any associated costs, especially in environments where productivity and efficiency gains are crucial. The lack of a strong cost barrier might also indicate that developers have access to affordable or company-provided LLM-oriented services, reducing the impact of cost perceptions on usage intentions. While prior research identified perceived cost as an obstacle to employing new technologies (Ahn & Park 2022), we suggest that cost concerns can be secondary to the direct benefits offered by LLMs in programming tasks. This aligns with several prior studies on AI-driven software adoption (Kim et al. 2023), where utility often outweighs financial concerns, particularly in professional settings where time efficiency and coding assistance are prioritized over monetary cost. However, the long-term impact of cost perceptions may evolve as LLM pricing models continue to change. In addition, further investigation into how perceived cost varies across different developer segments could present deeper insights into when and for whom cost becomes a more critical factor.

Our findings, while statistically robust, also offer important practical implications for both developers and tool creators. The observed significance of perceived usefulness and satisfaction in driving developers' continual usage intention translates into tangible benefits within specific software engineering contexts. For instance, 'satisfaction' with LLM-oriented programming tools can be directly linked to enhanced experiences in key developer activities. This includes the satisfaction derived from significantly reduced time spent on debugging complex or elusive errors, the ease and accuracy of generating comprehensive documentation for intricate codebases, streamlined and reliable refactoring suggestions, or the automated creation of robust unit and integration test cases that effectively cover

code functionalities. The 'usefulness' of LLM tools in practice extends beyond generic AI benefits, manifesting in specific ways that augment a developer's daily workflow. This encompasses improved code accuracy and reduced incidence of bugs, a diminished need for time-consuming manual searches on external platforms like Stack Overflow due to readily available context-aware suggestions, seamless integration of LLM functionalities directly within the Integrated Development Environment (IDE) to maintain flow state, acceleration of repetitive and boilerplate coding tasks, and valuable assistance in quickly exploring and understanding new APIs, libraries, or frameworks. By providing these concrete benefits, LLM tools not only boost individual developer productivity but also contribute to overall project efficiency and code quality.

Based on the findings of the current study, developers are required to prioritize LLM tools that fundamentally align with their expectations for task performance and usability in their daily workflow. This includes tools offering high code accuracy, intelligent context-aware suggestions that reduce the need for external search (e.g., Stack Overflow), and seamless integration within their preferred IDE to minimize disruption to their flow state. Prioritizing tools that consistently deliver on these fronts will enhance perceived usefulness and satisfaction, leading to sustained engagement.

Moreover, designers of LLM-oriented programming tools should pay attention to developing intuitive interfaces and robust feedback mechanisms for enhancing user confirmation and usability. This includes providing clear, actionable feedback on generated code, designing prompt engineering interfaces that are easy to learn, and implementing features that visibly demonstrate the tool's performance improvements (e.g., time saved, errors prevented). Such design choices are crucial for aligning with and exceeding user expectations, thereby promoting continual usage. In the case of engineering managers, they play a vital role in facilitating LLM tool adoption. They should manage the cost-effectiveness of these services by evaluating their return on investment in terms of increased developer productivity and reduced time-to-market. Furthermore, implementing comprehensive onboarding strategies and continuous training programs is critical. These initiatives should not only cover the technical aspects of using LLM tools but also address best practices for integrating them into existing development workflows and team collaboration, fostering an environment where developers feel supported in leveraging these new technologies.

Given the increasing utilization of LLM tools, educators are required to strategically align these tools with their educational goals. For novice programmers, this means integrating LLM-assisted coding into curricula to help them understand fundamental concepts, explore different coding solutions, and accelerate learning while still emphasizing critical thinking and independent problem-solving skills. Educators can guide students on how to effectively utilize LLMs as learning aids, focusing on understanding the generated code rather than merely accepting it, thereby preparing them for the evolving landscape of professional software development.

Beyond the direct statistical findings, we present some strategic directions LLM-oriented programming tools. Our study suggests that organizations should integrate user-centric evaluation criteria, derived from models like TAM and ECT, into their LLM tool assessment processes during pilot testing or procurement. Beyond technical benchmarks, evaluation should include systematic assessment of perceived usefulness (e.g., through surveys on productivity gains, task completion efficiency), perceived ease of use (e.g., learnability, cognitive load), initial satisfaction, and the extent to which the tool confirms or exceeds

developers' expectations. This approach ensures that purchased or adopted tools are not only technically capable but also highly likely to be continually used and embraced by the development teams.

While we present valuable insights into the adoption of LLM-oriented services, we did not explicitly account for counterfactuals or alternative explanations for the observed relationships. For instance, developers' prior exposure to AI-driven tools may also contribute to continued usage beyond the factors considered in our model.

This study also validated that perceived satisfaction is a notable mediator in the notable link between various antecedents and the intention to continually utilize LLM-oriented services (H9). This highlights that satisfaction, derived from positive interactions with the services, is a strong predictor of long-term usage intentions. Developers who are satisfied by the service performance tend to integrate them into their regular workflows, reaffirming the importance of user satisfaction in technology adoption.

Our findings on the critical role of perceived usefulness and satisfaction in driving continual usage intention resonate strongly with observed developer feedback and industry analyses of tools like GitHub Copilot and ChatGPT. For instance, the highly perceived usefulness reported by our participants aligns with studies indicating significant productivity gains for developers utilizing these AI assistants (e.g., Chen 2025; Chen et al. 2023, 2025), particularly in tasks involving boilerplate code generation, syntax correction, and initial problem-solving. Similarly, our emphasis on satisfaction, particularly related to expectation confirmation, reflects the importance of reliable and accurate AI outputs, a frequent topic in developer forums and user reviews concerning these tools.

Our empirical results can be directly interpreted through the lens of established industry frameworks such as SPACE (Satisfaction and Well-being, Performance, Activity, Communication and Collaboration, Efficiency and Flow) and DORA (DevOps Research and Assessment) metrics. The high 'satisfaction' among developers using LLM tools in our study directly contributes to the 'Satisfaction and Well-being' dimension of the SPACE framework, suggesting that positive user experience with these tools fosters a more content and potentially less stressed developer workforce. Furthermore, our findings on 'perceived usefulness', particularly its link to increased productivity, can be tied to DORA metrics like 'Lead Time for Changes' and 'Deployment Frequency.' By enhancing coding efficiency and reducing development cycles, LLM tools, when continually adopted, can contribute to faster software delivery and improved team performance, reflecting positively on these critical industry benchmarks.

As AI-assisted programming tools become increasingly integrated into software engineering practices, understanding the factors influencing their adoption is crucial. Our study contributes to this discussion by identifying key psychological and usability drivers behind LLM adoption, aligning with broader industry trends where AI-driven tools are reshaping software development workflows.

7 Conclusion

This research enhances our understanding of the factors influencing developers' acceptance and ongoing use of LLM-oriented services in programming tasks by integrating the ECT and TAM. The study's findings provide several important practical and theoretical implications.

From a theoretical viewpoint, the integration of ECT and TAM offers a robust framework for capturing both initial and sustained acceptance of LLM-oriented services in programming contexts. The results present the critical role of confirmation in mediating the effects of expectations on developers' usefulness and ease of use feelings. This confirms the centrality of satisfaction as a notable role of determining continual intention to use, consistent with the findings of prior technology adoption research (Talwar et al. 2020).

Practically, the findings emphasize key factors that technology developers and organizations should consider when promoting LLM-oriented programming tools. To enhance user satisfaction and encourage sustained adoption, it is crucial to ensure that these tools meet or exceed developers' expectations while emphasizing their ease of use and practical benefits. Addressing potential cost concerns by demonstrating the value and return on investment of LLM-oriented services may also help mitigate any perceived financial barriers.

The findings of this study present that expectation confirmation plays a main role in forming developers' ease of use, usefulness, and satisfaction with LLM-oriented services in programming tasks. Thus, we present several strategies for LLM designers: First, providing clear, and scenario-specific guidelines can help developers set realistic expectations and eliminate potential disappointment. Second, carrying out context-aware suggestions with interactive feedback can improve LLM adaptability to both programming styles and requirements of developers. Third, considering that perceived usefulness is one of the greatest motivations of developer acceptance, the tools should focus not only on improving efficiency but also on providing flexible pricing solutions to ensure accessibility.

To enable engineering teams to track the impact of LLM tools, we propose practical metrics for satisfaction and perceived usefulness. For satisfaction, teams could administer periodic brief surveys using scales like those in this study (e.g., overall satisfaction, meeting expectations), track Net Promoter Score (NPS) among LLM tool users, or analyse qualitative feedback for recurring positive themes. For perceived usefulness, quantifiable metrics might include observed reductions in time spent on specific coding tasks (e.g., debugging, documentation), the frequency of LLM-generated code adoption in pull requests, or developers' self-reported productivity gains through regular internal surveys. While specific benchmarks may vary by organization and tool, establishing internal baselines and tracking progress against these metrics can provide tangible evidence of the LLM tools' value and guide continuous improvement.

However, this study has limitations, including its focus on a specific context and region (South Korea), which can reduce the potential generalizability of the results to other cases or regions. Future research can explore cross-cultural differences in service acceptance or examine how various types of programming tasks influence perceived usefulness and ease of use. We also indicated that several fit indices are not excellent by the guidelines of several prior research (e.g., RMSEA, CFI; Hu and Bentler (1999)). In addition, because we conducted our pilot survey with graduate students, they may have different usage purposes, expectations, and challenges compared to industry professionals. Thus, future research can start their exploratory approach with more professional workers. Furthermore, we did not pay notable attention to both types and flavors of LLM services. While we can provide several insights, future research should examine whether different functionalities and types of services influence user perceptions. In addition, we used SEM to examine a series of

relationships in the research model. However, SEM mainly identifies statistical connections rather than definitive causal relationships. Thus, future studies should examine counterfactual analysis techniques, such as experimental validation, to improve causal claims. In addition, we present several empirical insights into LLM adoption using SEM, the insights depend on a self-reported survey. Thus, future research is required to verify these insights via qualitative approaches, such as developer interviews.

While we employed the constructs affecting the intention, it mainly focuses on positive aspects. However, it does not explicitly account for potential negative factors. Future research should consider several comprehensive scenarios, which can be examined in programming tasks. In addition, because prior research indicated that diverse dimensions of perceived satisfaction can be addressed in user behavior (Dou et al. 2021), it is required to address the dimensions of satisfaction in providing a better understanding of developers' intention to use LLMs.

In the case of perceived cost, we did not account for potential temporal fluctuations. As technological advancements lead to cost reductions in LLMs, the perceived cost may change over time. This temporal variability could influence the validity of our findings, particularly in terms of the relationship between perceived cost and LLM adoption. Therefore, longitudinal approaches could be valuable for gaining deeper insights into how developers' perceptions evolve and how these perceptions shape their views toward such services. Moreover, the finding presented in the current study that perceived cost is one of the insignificant factors addressing the intention. The majority of developers in our South Korean sample likely had their subscription fees for such services covered by their respective companies. This particular finding may therefore not directly generalize to contexts where individual developers are solely responsible for the cost of LLM-oriented services, suggesting a potential limitation in the broader applicability of our cost-related conclusions.

As one of the additional limitations, we conducted a survey at a single point in time, inherently limits our ability to capture the long-term evolution of developer acceptance of LLM-oriented services. Future research employing longitudinal approaches would offer a more dynamic understanding of these adoption patterns over extended periods. Moreover, we employed Harman's single-factor test as an initial diagnostic. We acknowledge, in line with recent critiques (e.g., Howard et al. 2024), that this test is a preliminary and often insufficient method for definitively ruling out CMB. Its inability to reliably distinguish between research designs that minimize or are vulnerable to CMB suggests that while our survey demonstrated internal consistency and acceptable model fit, the presence of some degree of common method variance cannot be entirely dismissed by this test alone. More robust methods, such as the use of marker variables or procedural remedies, would be beneficial in future research to further mitigate and assess CMB.

A further methodological consideration pertains to measurement equivalence. While rigorous back-translation procedures and expert review were employed to ensure linguistic equivalence of our survey instruments, formal testing for measurement invariance (e.g., configural, metric, or scalar invariance) across different linguistic or cultural groups was not conducted in the current study. Consequently, we cannot definitively ascertain whether the constructs operate equivalently in the translated context, which could potentially intro-

duce concerns about measurement errors and limit direct cross-cultural comparisons. Future research should explicitly address this limitation by conducting robust measurement invariance testing to ensure the stability and comparability of the measurement model and its underlying constructs across diverse populations.

Furthermore, while our model exhibited acceptable fit statistics, concerns regarding potential overfitting remain. Specifically, residual plots were not formally assessed for heteroscedasticity, an important diagnostic for ensuring the consistency of error variance across all levels of the independent variables. Additionally, although bootstrapping was utilized to evaluate the in-sample reliability and stability of our coefficient estimates, a formal train/test split with k-fold cross-validation was not performed to explicitly assess model performance on unseen data. While our large sample size (N = 1,338) contributes to statistical power, the absence of these external validation techniques means the generalizability and predictive accuracy of our model to new, unobserved datasets should be interpreted with caution. Future research should prioritize these rigorous validation methods, including thorough residual analysis and cross-validation, to enhance the robustness and external validity of findings in this domain.

Lastly, a methodological consideration relates to the design of our Likert scale. While participants were allowed to skip items, we acknowledge that the use of 'Neutral' as the midpoint on our 7-point scale (rather than 'Neither agree nor disagree') could introduce interpretive ambiguity. In latent variable modeling, 'Neutral' can ambiguously reflect indifference, disengagement, or confusion, potentially leading to inflated variance, biased factor loadings, and misestimated path coefficients, as it is numerically positioned more favorably than 'Disagree' without necessarily reflecting a clearer attitudinal stance. Future studies should consider adopting more explicit midpoint phrasings, such as 'Neither agree nor disagree,' to ensure clearer attitudinal representation and enhance the robustness of measurement in similar models.

In summary, this study advances our knowledge of the motivational constructs driving the acceptance and continued use of LLM-oriented services in programming tasks. It offers valuable insights for both academic researchers and industry stakeholders seeking to enhance the adoption of AI-driven programming assistants.

As the software engineering landscape continues its rapid evolution with the widespread adoption of generative AI, our findings highlight key areas for future focus. The alignment of our results with public developer feedback underscores a shared understanding of LLMs' potential and current challenges. Future research, integrating real-world usage logs and A/B testing with diverse developer populations, will be crucial to further refine our understanding and provide even more robust insights that directly inform the next generation of AI-powered development tools and industry best practices.

Moreover, although our focus group interviews provided valuable practitioner validation for our model's cornerstones, future research could explore broader qualitative methodologies, such as longitudinal case studies or observational research across diverse organizational settings, to capture more nuanced behavioral insights over time and across different project lifecycles.

Appendix A.

Table 9 Questionnaire items in the survey

Construct	Items
Expectation (Gupta et al. 2020; Eren 2021)	EXP1: "*I expected LLM-oriented services to be useful for programming tasks.*" EXP2: "*I expected LLM-oriented services to improve my programming task convenience.*" EXP3: "*I expected LLM-oriented services to enable me to make programming tasks quickly*." EXP4: "*I expected my interaction with LLM-oriented services in programming tasks to be clear and understandable.*"
Confirmation (Park 2020; Jumaan et al. 2020)	CON1: "*My experience with LLM-oriented services in programming tasks is better than what I expected*." CON2: "*Overall, the majority of my expectations from using LLM-oriented services in programming tasks are confirmed*." CON3: "*The functionalities of LLM-oriented services in programming tasks meet my expectations*."
Perceived Ease of Use (Rajak & Shaw 2021; Zhong et al. 2021)	PEOU1: "*My interaction with LLM-oriented services in programming tasks is clear*." PEOU2: "*Using LLM-oriented services in programming tasks is easy for me*." PEOU3: "*Interacting with LLM-oriented services in programming tasks does not require mental effort*."
Perceived Usefulness (Rajak & Shaw 2021; Jumaan et al. 2020)	PU1: "*I think that LLM-oriented services in programming tasks are useful for my work*." PU2: "*Using LLM-oriented services in programming tasks increases my productivity*." PU3: "*Using LLM-oriented services in programming tasks helps me conveniently perform many sub-tasks.*"
Satisfaction (Kim et al. 2021; Park et al. 2016)	SAT1: "*Overall, I am satisfied with LLM-oriented services in programming tasks.*" SAT2: "*The LLM-oriented services in programming tasks I am currently using meet my expectations*." SAT3: "*I am very pleased with my experience with using LLM-oriented services in programming tasks.*"
Perceived Cost (Park 2019; Jang & Park 2019)	PC1: "*There are financial barriers to using LLM-oriented services in programming tasks*." PC2: "*I think that LLM-oriented services in programming tasks are expensive.*" PC3: "*Overall, using LLM-oriented services in programming tasks costs me a lot of money.*"
Continual Intention to use (Davis 1993; Gupta et al. 2020)	CIU1: "*I intend to use LLM-oriented services in programming tasks as much as possible.*" CIU2: "*I intend to continually use LLM-oriented services in programming tasks.*" CIU3: "*I recommend that other consumers use LLM-oriented services in programming tasks.*"

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Data availability All dataset in the current study is available at https://github.com/dxlabskku/

Declarations

Ethical approval This study was approved by the Institutional Review Board (IRB) of Sungkyunkwan University (2024–05-043).

Informed consent Participants had to complete an informed consent form before beginning the survey.

Clinical trial number Not applicable.

Conflict of interest The author declared that they have no conflict of interest.

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